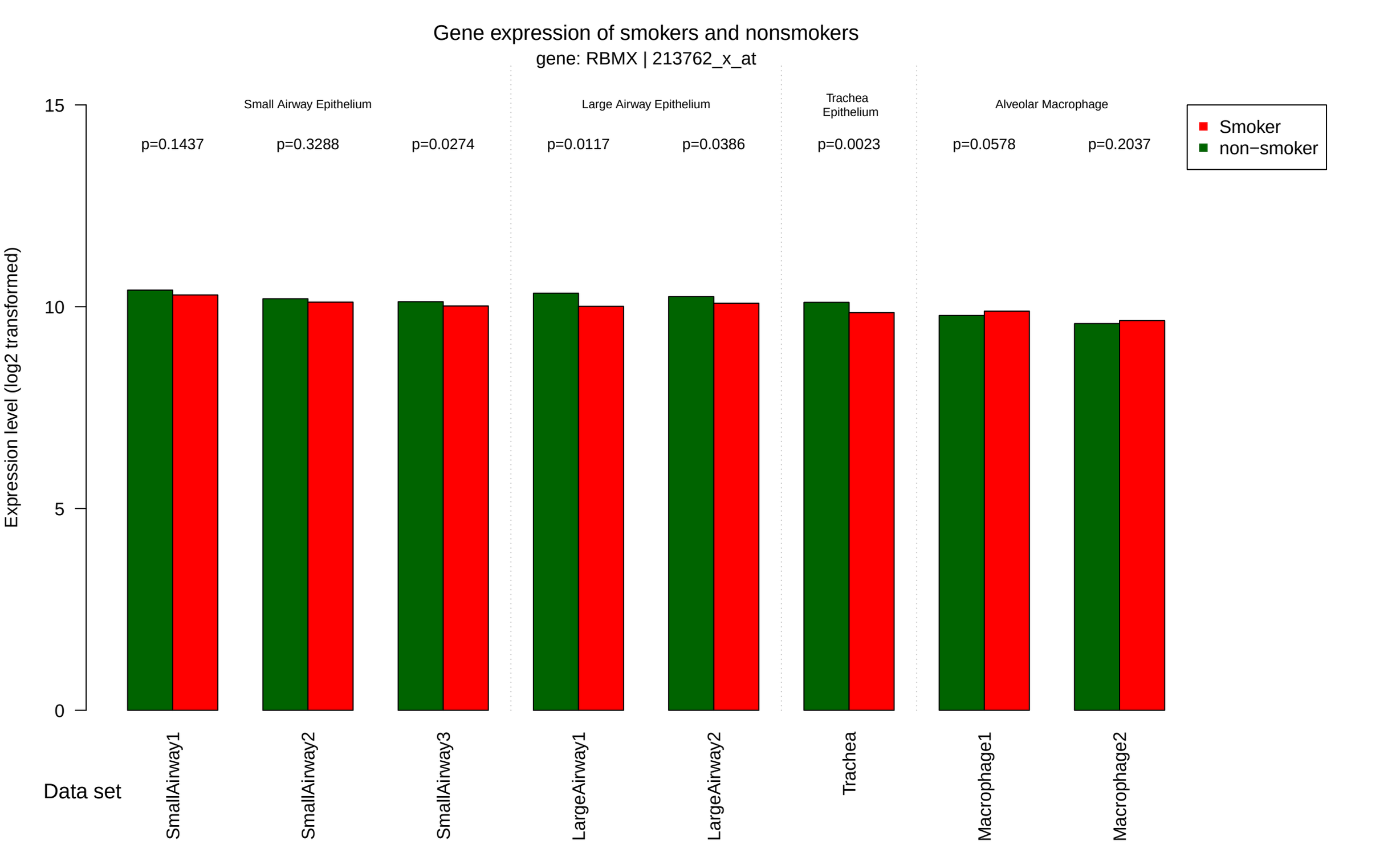
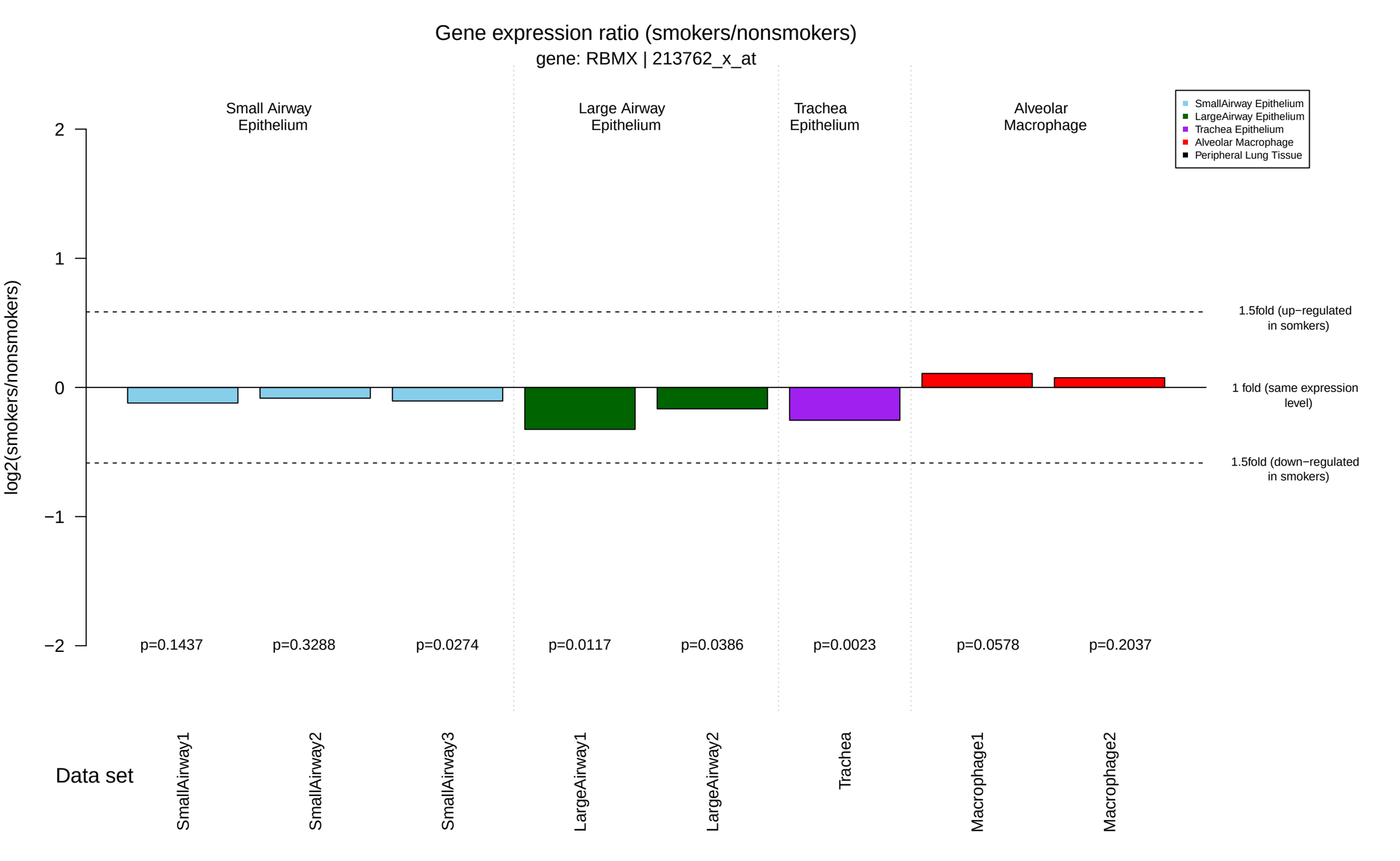
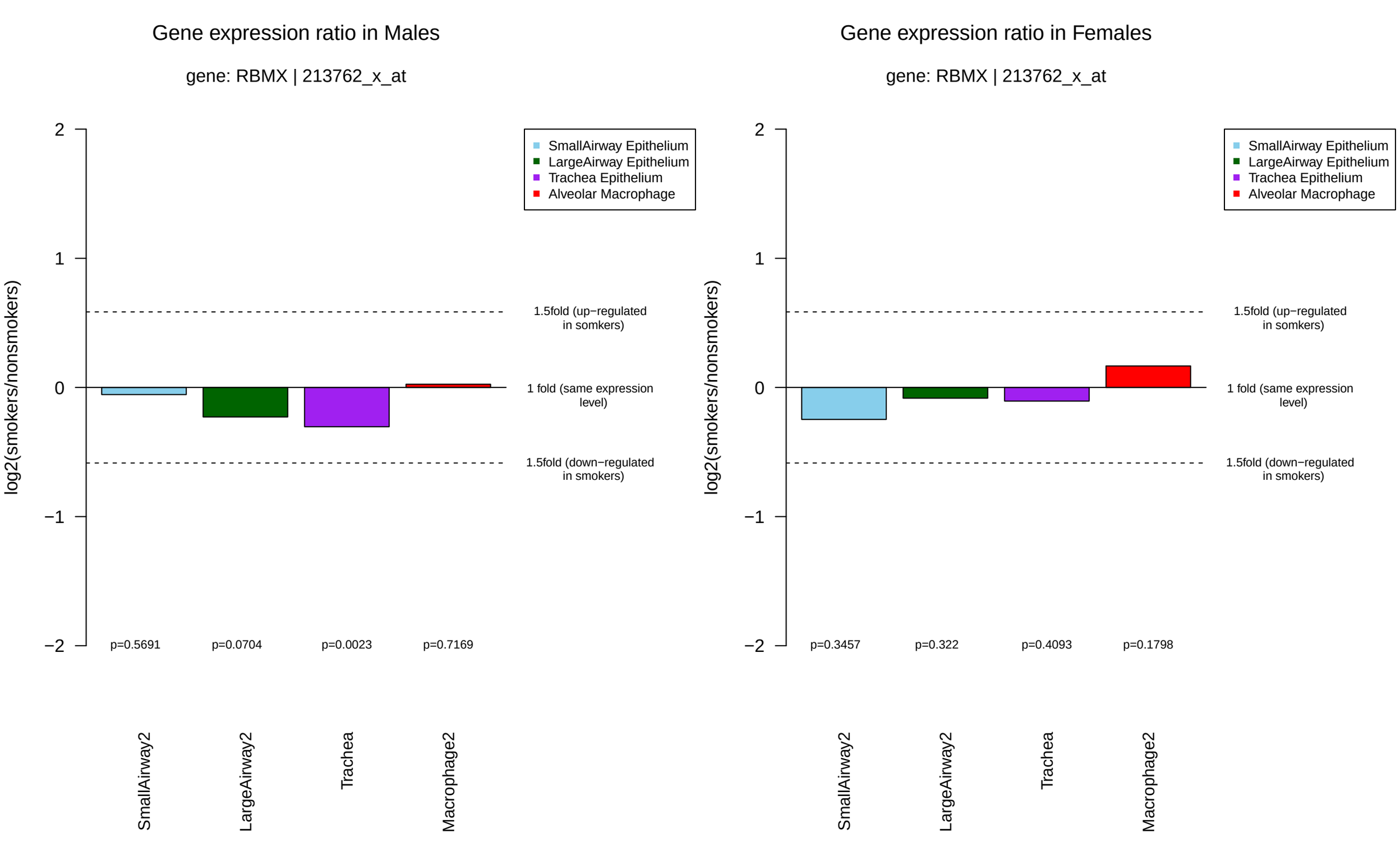
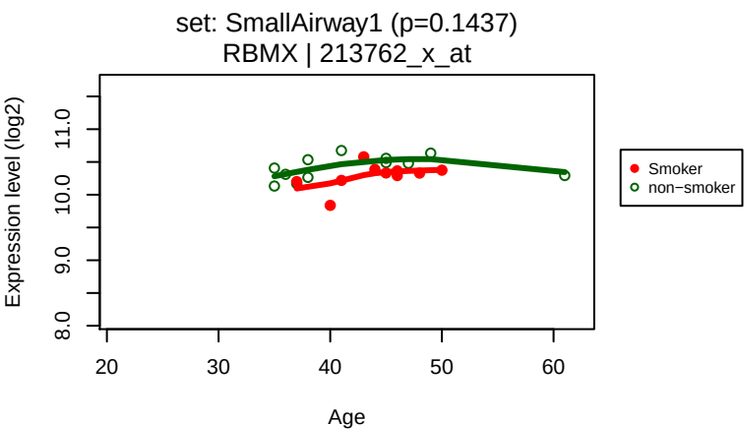
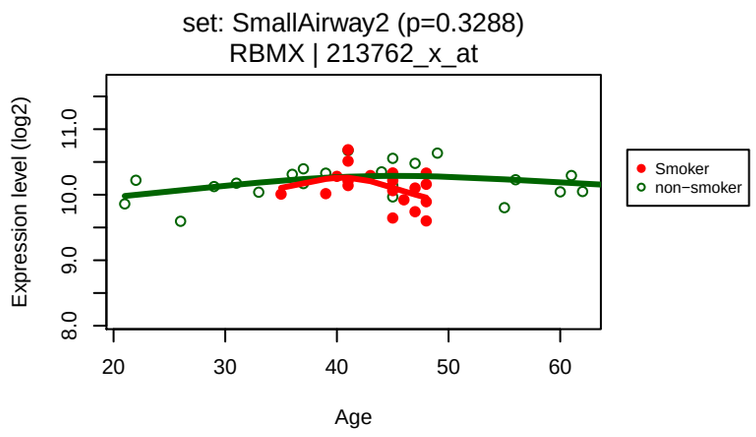
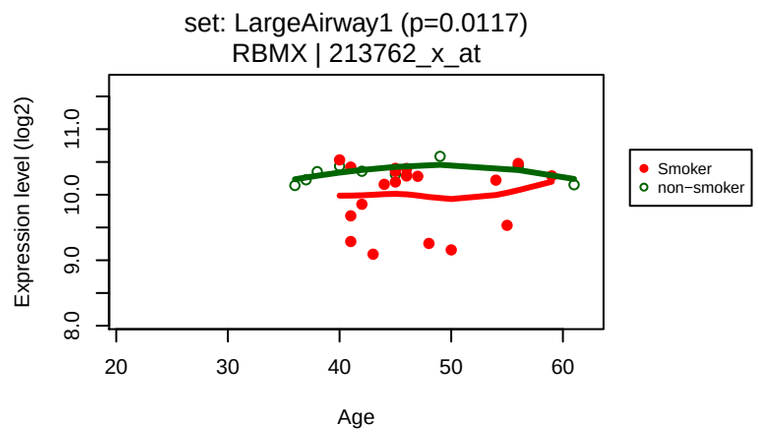

Gene expression of smokers and nonsmokers
gene: RBMX | 213762_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1437
p=0.3288
p=0.0274
p=0.0117
p=0.0386
p=0.0023
p=0.0578
p=0.2037
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

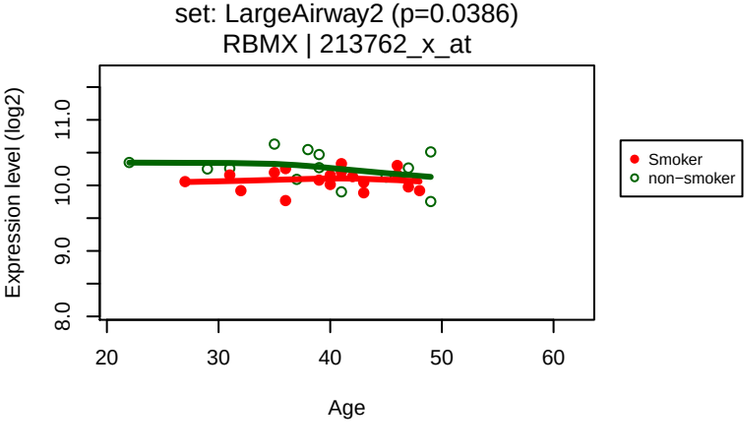

Gene expression ratio (smokers/nonsmokers)
gene: RBMX | 213762_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1437
p=0.3288
p=0.0274
p=0.0117
p=0.0386
p=0.0023
p=0.0578
p=0.2037
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

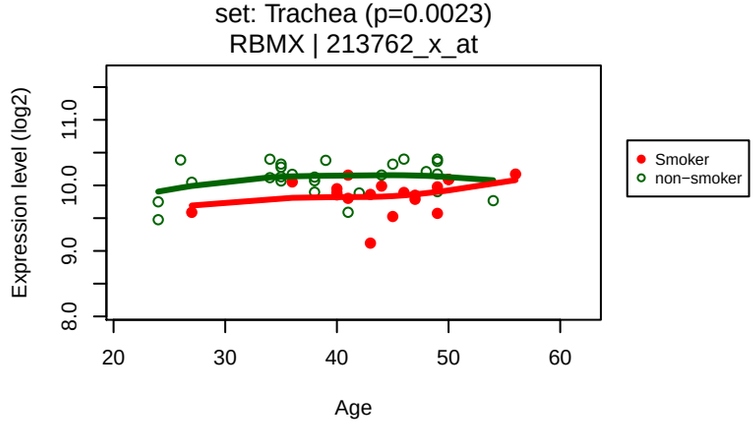

Gene expression ratio in Males
gene: RBMX | 213762_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5691
p=0.0704
p=0.0023
p=0.7169
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBMX | 213762_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3457
p=0.322
p=0.4093
p=0.1798
SmallAirway2
LargeAirway2
Trachea
Macrophage2

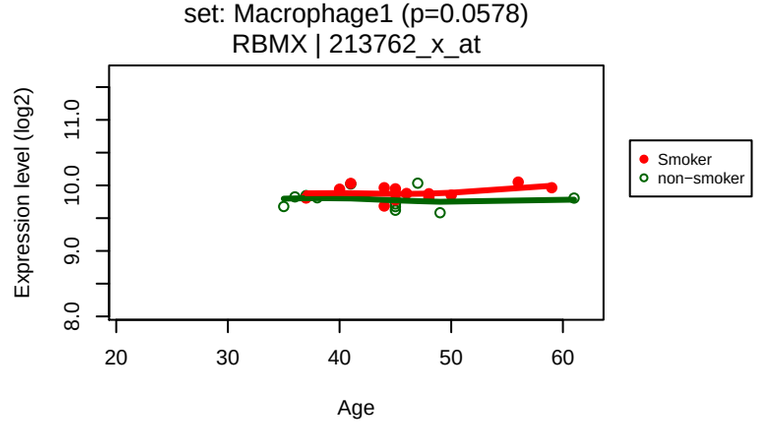

set: SmallAirway1 (p=0.1437)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker

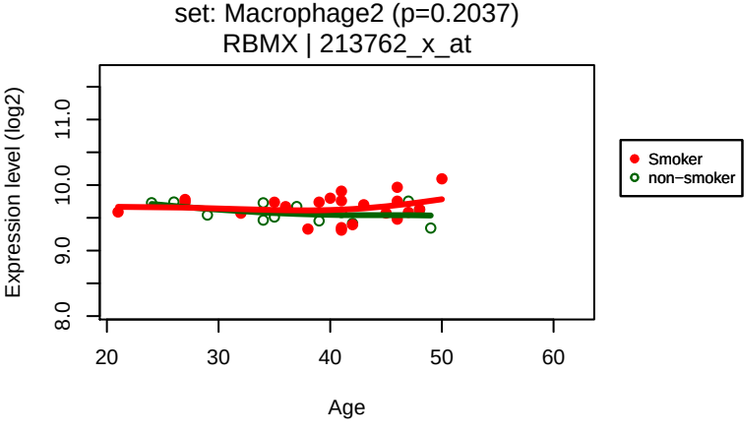

set: SmallAirway2 (p=0.3288)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker

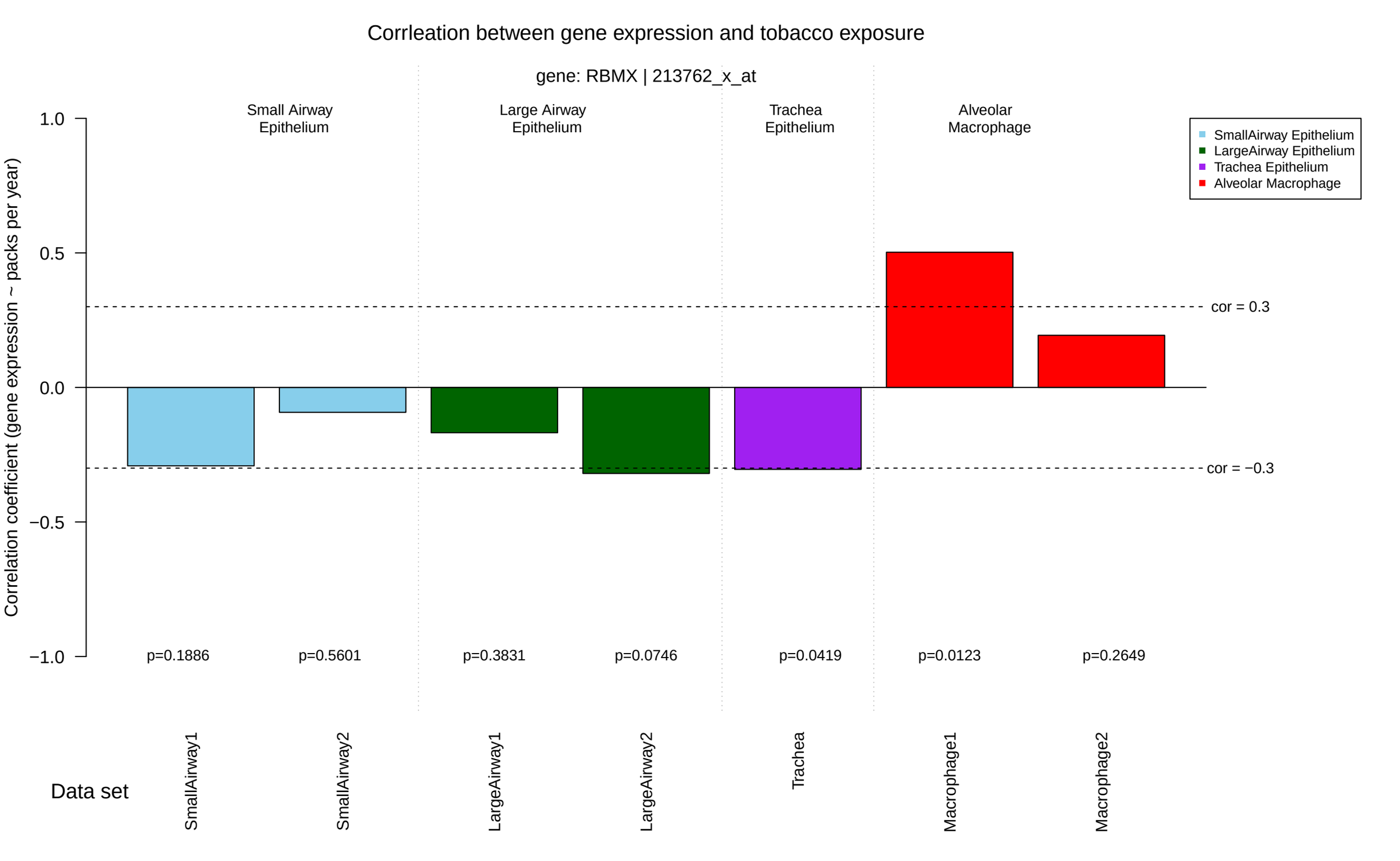

set: LargeAirway1 (p=0.0117)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0386)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0023)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0578)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2037)
RBMX | 213762_x_at
Expression level (log2)
8.0
9.0
10.0
11.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBMX | 213762_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1886
p=0.5601
p=0.3831
p=0.0746
p=0.0419
p=0.0123
p=0.2649
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2